

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2455854.73649 +/- 0.0004 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1684 +/- 0.0056
Transit depth (Rp/Rs)^2	2.83 +/- 0.19 [%]
Orbital Inclination (inc)	89.36 +/- 0.54
Airmass coefficient 1 (a1)	1.0507 +/- 0.009
Airmass coefficient 2 (a2)	-0.044 +/- 0.017
Scatter in the residuals of the lightcurve fit is	0.86 %
Best Comparison Star	#3 - [238, 232]
Optimal Aperture	3.65
Optimal Annulus	10.0
Transit Duration (day)	0.095 +/- 0.0013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1684 ± 0.0056 vs 0.1592 (deviation 1.17σ)
Duration fit vs published	0.095 d vs 0.0946 d (deviation 0.23σ)
Mid-time O–C	-3.0 min
Depth SNR	21.4
Residual scatter	0.86 %

Light curve

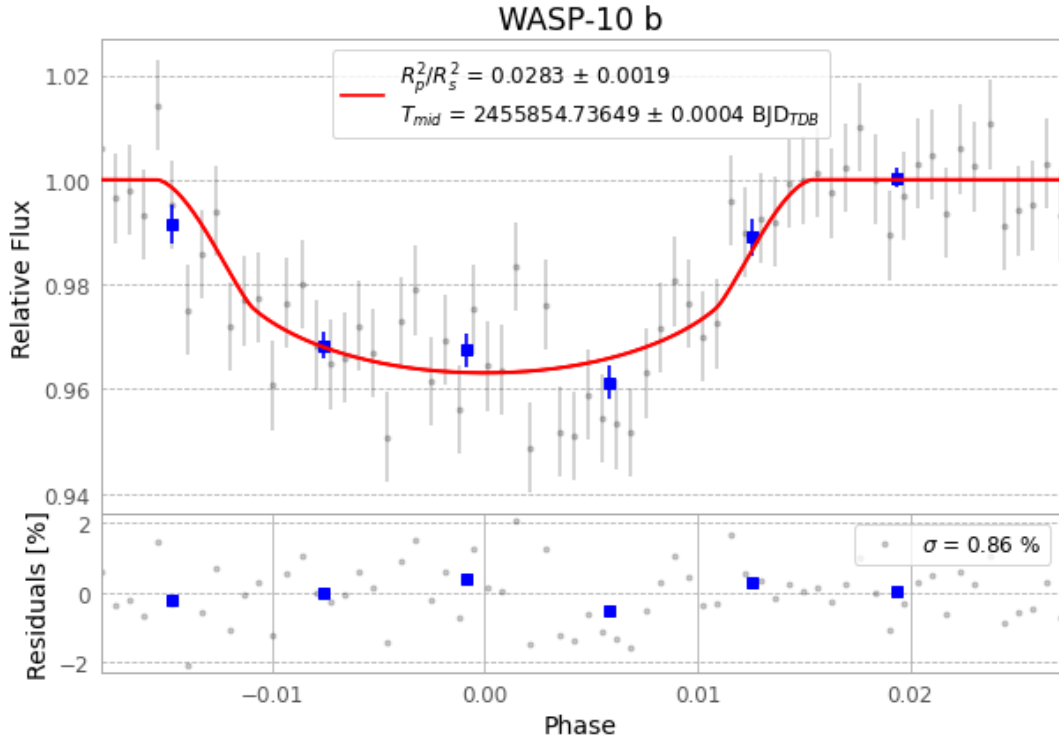


Figure 1 — FinalLightCurve_WASP-10 b_2011-10-19.png (SHA-256 8d9f694438d20b96...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [297, 277], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8fb42a1174950887...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

68 FITS frames (45 MB), WASP-10111020041214.FITS → WASP-10111020073312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-111020.log (793e0e6cf7d8...), FinalLightCurve_WASP-10 b_2011-10-19.png (8d9f694438d2...), FinalLightCurve_WASP-10 b_2011-10-19.csv (0577a697b2b2...)

Compute resources

Wall time 144.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 00:15Z.